

Paper Corpus Generation: An End-to-End Account

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Abstract

This note follows one case from a random seed to the arrays read by the trainer. The replacement corpus contains four kinds of periodic water-wave states: Stokes waves, Tanaka-profile sums, Benjamin–Feir perturbations, and JONSWAP/TMA random seas. Every label is computed with the same fixed discrete Dirichlet–Neumann operator.

A draw must first lie in the stated parameter range of its family. A time-dependent draw is then retained only if its declared fixed-step calculation returns a complete finite trajectory and keeps the free surface above the bottom. Benjamin–Feir revision 4 and JONSWAP/TMA revision 4 trajectories must also conserve their full internal-band Hamiltonian to relative error at most 10^{-3} over the autonomous saved-time grid. No rule based on visual roughness, sign changes, slope, or model error removes a case. This document specifies the replacement corpus. The source-authenticated release is complete at 16384 training, 1024 validation, and 1024 test cases per family: 73728 accepted cases from 74045 attempts, 317 zero-row rejections, and 7686144 retained rows. No reported model has been trained on this release.

1 Objects and guiding rules

The horizontal interval is $0 \leq x < L$, with periodic endpoints and $L = 2\pi$, and gravity is set to one. A *case* is one sampled initial surface elevation $\eta_0(x)$, one sampled value $\xi_0(x)$ of the velocity potential at the surface, and one fixed water depth $h > 0$. A *trajectory* is the computed evolution of that case. A *row* is one retained time from an accepted case.

The case, rather than the row, is the unit of splitting, rejection, and training weight. This matters because an accepted Stokes case stores one row, whereas accepted Tanaka and Benjamin–Feir cases each store 200 rows. Counting all stored rows equally would therefore give a long trajectory much more influence than a static wave.

The construction follows four elementary rules.

1. Fix the numerical meaning of a label before sampling any cases.
2. Assign a case to training, validation, or test before seeing whether its numerical calculation succeeds.
3. Replace a failed attempt only within its original parameter category.
4. Select stored times only after the entire case has been accepted.

2 The common numerical label

For given η , ξ , and h , let $\varphi(x, y)$ solve

$$\varphi_{xx} + \varphi_{yy} = 0, \quad \varphi_y(x, -h) = 0, \quad \varphi(x, \eta(x)) = \xi(x). \quad (1)$$

Subscripts denote partial derivatives. The physical Dirichlet–Neumann operator is

$$G(\eta; h)\xi = (\varphi_y - \eta_x \varphi_x)_{y=\eta(x)}. \quad (2)$$

It is the slope-corrected upward derivative of the solution φ at the free surface. In the Zakharov equations it is also η_t , the instantaneous rate of change of the surface.

The dataset does not claim to evaluate (2) exactly. Instead, it freezes one discrete reference calculation. All fields use $N = 1024$ equally spaced points $x_\ell = 2\pi\ell/1024$, $\ell = 0, \dots, 1023$. Since the period is 2π , the Fourier wavenumbers are the integers. Let P_{128} discard all modes with $|k| > 128$, and let P_{128}° additionally remove the

spatial mean. Let $\mathcal{G}_6^{1024,8}$ denote the implemented order-six Craig–Sulem calculation on 1024 points with padding factor eight. The label is

$$\boxed{G_{\text{ref}}(\eta; h)\xi = P_{128}^{\circ} \left[\mathcal{G}_6^{1024,8}(P_{128}\eta; h)(P_{128}\xi) \right]}. \quad (3)$$

The calculation is performed in 64-bit arithmetic. Only the final spatial arrays are converted to 32-bit form for storage. Thus every retained row is

$$(P_{128}\eta(t), P_{128}\xi(t), h, t, G_{\text{ref}}(\eta(t); h)\xi(t)). \quad (4)$$

The model is trained to reproduce the fixed discrete quantity in (3). Approximation of the continuum operator (2) is a separate question requiring convergence tests in truncation order, grid size, and padding. Every constructor removes the spatial mean of ξ ; this fixes the irrelevant freedom to add a constant to the potential.

JONSWAP/TMA is evolved on 2048 points, whereas every stored row has 1024 points. For an N' -point field, define the normalized Fourier coefficient

$$\hat{f}_k^{(N')} = \frac{1}{N'} \sum_{j=0}^{N'-1} f\left(\frac{2\pi j}{N'}\right) e^{-2\pi i j k / N'}. \quad (5)$$

At a retained JONSWAP/TMA time, the program copies the coefficients of η and ξ with $|k| \leq 128$ from the 2048-point grid, reconstructs those two fields on 1024 points, and evaluates (3) there. It never resamples the internal $G(\eta; h)\xi$: the stored target is recomputed because it is defined by the 1024-point order-six calculation.

3 One case from beginning to end

The complete production path is:

1. Assign a family, family revision, parameter category, data split, and deterministic random-number key, then draw and record all initial-condition parameters.
2. Check the family’s stated parameter rules and construct (η_0, ξ_0, h) . Before rollout, the assembled initial arrays must be finite, $h > 0$, and $h + \eta_0 > 0$ on the grid.
3. For a Stokes case, evaluate (3) at $t = 0$. For Tanaka and Benjamin–Feir, run the autonomous fixed-step calculation in Section 5. For JONSWAP/TMA, first run its declared nonlinear adjustment, hand its full internal-band endpoint to the autonomous solver, and restart the production clock at zero.
4. Apply the whole-case decision in Section 5. A failed trajectory contributes no rows, including no apparently good prefix.
5. After acceptance, choose the retained times and write their arrays, numerical diagnostics, and checksum.
6. Count an accepted case toward its original parameter category. If it failed, draw a new case in that same category.
7. When every family and split quota is complete, build a checked index into the array files and pass that index to the trainer.

There are only two case-level gates: membership in the family’s declared construction support, followed by a complete admissible calculation on the requested interval. A declared support failure contributes zero rows. It is distinct from an unexpected construction or implementation error, which stops the run. The sampled specification is written before numerical work, so an interrupted case can be replayed without drawing a replacement. An unexpected construction error, including failure of the initial-array conditions above, stops the run; it is not silently relabeled as an ordinary rejected draw. The declared JONSWAP/TMA phase-realization check is the exception: nonfinite assembled fields or a nonpositive realized water column produce a recorded zero-row construction rejection for that case, which is replaced in the same category while valid batch companions continue. Other unclassified constructor failures still stop the run. Checksums detect missing or changed array files.

4 How the four initial-condition families are sampled

A *parameter category* is a pre-set part of one family’s parameter range that receives its own accepted-case count. The categories keep important regimes from disappearing by chance. Uniform sampling on $[a, b]$ means that every

interval of the same length is equally likely. Log-uniform sampling means that $\log h$ is uniform. A uniform simplex split of a positive number A chooses nonnegative weights w_j with $\sum_j w_j = 1$ and sets the pieces to Aw_j . Unless an endpoint convention is written explicitly, closed brackets describe the closure of the continuous support. The ordinary floating-point uniform draw includes its lower endpoint and excludes its upper endpoint. Explicit endpoint conventions in an individual sampling formula—including half-open phase and center intervals and the mixed open–closed Benjamin–Feir steepness draw—override that default. The distinction has probability zero for the continuous law but is recorded for exact replay.

Table 1: The four training families. A horizon is needed only for a time-dependent family.

Family	Categories	Main distinction	Horizon / retained rows
Stokes	4	finite/deep branch and low/moderate steepness	static / 1
Tanaka	11	regime, number of crests, and number moving right	$T = 200$ / 200
Benjamin–Feir	66	one feasible carrier–sideband integer pair per category	last 0.08-grid time not exceeding $100T_c$ / 200
JONSWAP/TMA	27	shallow/finite/deep, peak enhancement, and direction balance	last 0.08-grid time not exceeding 16 peak periods / 16

4.1 Fifth-order Stokes states

A Stokes draw first fixes one of four parameter categories:

$$\{\text{finite, deep}\} \times \{0.005 \leq ka < 0.03, \quad 0.03 \leq ka \leq 0.15\}.$$

Here k is the carrier wavenumber and a is the expansion-amplitude parameter of the fifth-order formula. Because the first harmonic receives nonlinear corrections, a is not exactly the realized first-harmonic amplitude. The common parameter interval is

$$0.000766 \leq a \leq 0.011494.$$

For the finite-depth branch, the mode number is $n \in \{14, \dots, 26\}$, the wavenumber is $k = n$, and

$$0.02 \leq h \leq 1.5, \quad 0.5 \leq kh \leq 5.$$

For the deep branch,

$$n \in \{1, \dots, 20\}, \quad 4 \leq h \leq 50, \quad kh \geq 5.$$

The branch is assigned before n and h are drawn. Thus the shared boundary $kh = 5$ does not select a formula: a finite-category case uses the finite-depth coefficients and a deep-category case uses the deep-water coefficients. Allowing equality in both branch-specific supports therefore creates no ambiguous case record. The program lists the modes compatible with the chosen category and selects one uniformly. It then draws h log-uniformly in the compatible depth interval, draws a phase ϕ uniformly on $[0, 2\pi)$, and draws a uniformly from the intersection of the common amplitude interval and the chosen ka interval. The code evaluates Fenton’s fifth-order analytic Stokes expansion, using its finite-depth coefficients or deep-water limit, to construct (η_0, ξ_0) [1]. The parameter ranges, four parameter categories, random phase, and depth law above are corpus choices rather than claims made by Fenton.

The finite-depth formula is used only in its intended Stokes regime. For phase $\theta = kx + \phi$, write its elevation as

$$\eta_5(\theta) = \sum_{r=1}^5 E_r \cos(r\theta),$$

where E_r is the coefficient of harmonic r . Let H_{4096} be the largest sampled value of η_5 minus its smallest sampled value on 4096 equally spaced phases, and define

$$H_+ = H_{4096} + \frac{1}{4} \left(\frac{2\pi}{4096} \right)^2 \sum_{r=1}^5 r^2 |E_r|.$$

The last term bounds the possible height missed between two phase samples. With wavelength $\lambda = 2\pi/k$, the required support condition is

$$\text{Ur}_+ = \frac{H_+ \lambda^2}{h^3} \leq 26. \quad (6)$$

This is a conservative Ursell-number restriction for the fifth-order formula, not a fitted predictor of whether a numerical rollout would survive [1, 2]. If (6) fails, the mode, depth, phase, and parameter category remain fixed and only the amplitude is redrawn, with at most 1000 redraws. Exhausting that limit gives a zero-row attempt, and the category receives a new case. A valid Stokes case is static: it stores only (4) at $t = 0$.

4.2 Periodic Tanaka-profile sums

The three main Tanaka structural groups contain $m = 1, 2, 3$ component profiles. First draw their total dimensionless amplitude

$$A \sim \text{Unif}[0.10, 0.35].$$

Next draw $(W_1, \dots, W_m) \sim \text{Dirichlet}(1, \dots, 1)$ and set

$$\alpha_j = AW_j, \quad j = 1, \dots, m.$$

For $m = 1$, $W_1 = 1$. Hence α_j is the amplitude-to-depth ratio of component j and $\sum_j \alpha_j = A$.

Depth is drawn after the amplitudes. Define the largest component ratio and its inverse-width proxy by

$$\alpha_{\max} = \max_j \alpha_j, \quad \kappa_{\max} = \frac{\sqrt{3\alpha_{\max}}}{2h}. \quad (7)$$

The expression for $\kappa_{\max}$ is the inverse-width scale in the small-amplitude KdV solitary-wave formula. We use it only as an elementary resolution proxy for the narrowest requested crest; it is not the exact width of a finite-amplitude Tanaka wave. Since the delivered band ends at $|k| = 128$, define

$$R = \frac{128}{\kappa_{\max}}. \quad (8)$$

The revision-3 support requires $R \geq 10$. Equivalently, set

$$h_{\min} = \max \left\{ h_{\text{base}}, \frac{10\sqrt{3\alpha_{\max}}}{256} \right\}, \quad h = h_{\min}^{1-U_h} h_{\max}^{U_h}, \quad U_h \sim \text{Unif}[0, 1]. \quad (9)$$

The main groups use $h_{\text{base}} = 0.01$ and $h_{\max} = 0.30$. The steep structural group has one crest and uses $A = \alpha_1 \sim \text{Unif}[0.25, 0.45]$, $h_{\text{base}} = 0.20$, and $h_{\max} = 0.35$. For every steep amplitude, the second entry in the maximum in (9) is less than 0.20. Thus the conditional lower bound is inactive there and the steep depth remains the geometric interpolation from 0.20 to 0.35.

At fixed amplitudes, uniform U_h makes $\log h$ uniform on the declared interval. This gives equal increments of multiplicative profile scale; it is a numerical coverage rule, not a probability model for physical depths.

Let $s_j \in \{-1, +1\}$ denote the direction of crest j , with $s_j = +1$ for rightward travel, and let

$$q = \#\{j : s_j = +1\}$$

be the number of right-moving crests. For a main case, each pair (m, q) with $m \in \{1, 2, 3\}$ and $q = 0, \dots, m$ is a parameter category. These give $2 + 3 + 4 = 9$ categories. The steep one-crest group contributes the two categories $q = 0, 1$, for eleven Tanaka categories in total. Conditional on q , the sampler chooses uniformly which q of the m exchangeable crest labels move right; the rest move left.

The accepted-case quota is divided as evenly as possible across all eleven categories. Therefore the distinct direction compositions receive equal coverage, rather than the binomial frequencies generated by independent fair signs. Left and right directions remain balanced marginally in the ideal equal-category population. For a finite quota, category counts differ by at most one and a one-case direction imbalance is possible; the fixed order gives a quota of 1024 one extra left-moving one-crest case. Because the one-, two-, and three-crest main groups contain two, three, and four direction compositions, their aggregate shares are $2/11$, $3/11$, and $4/11$; the steep group has share $2/11$. This deliberately replaces the old four-group independent-sign mixture.

The direction categories were introduced in Tanaka revision 2. Revision 3 retains all eleven categories but draws amplitudes before depth and uses (9). A revision-2 proposal remains exactly replayable, but it cannot be resumed into a revision-3 run. Benjamin-Feir and JONSWAP/TMA use revision 4, and static Stokes remains at revision 2.

The value 10 was fixed by a full-horizon boundary calculation with $\alpha \in \{0.225, 0.30, 0.35\}$, $R \in \{6, 8, 10\}$, and internal cutoffs $J \in \{224, 256\}$. For the cutoff- J history, define

$$d_J(t) = P_{128}^\circ \left[\mathcal{G}_6^{1024,8}(\eta_J(t); h) \xi_J(t) \right] - G_{\text{ref}}(P_{128}\eta_J(t); h) P_{128}\xi_J(t)$$

and

$$C_J = \frac{\max_t \|d_J(t)\|_{2,N}}{\max_s \|G_{\text{ref}}(P_{128}\eta_J(s); h) P_{128}\xi_J(s)\|_{2,N} + 10^{-12}},$$

where $\|\cdot\|_{2,N}$ is the root-mean-square norm on the spatial grid. The reported statistic is $\max_{J \in \{224, 256\}} C_J$. Its worst values were 2.91×10^{-3} at $R = 6$, 6.31×10^{-4} at $R = 8$, and 1.40×10^{-4} at $R = 10$. Thus $R = 6$ exceeded the 10^{-3} budget, whereas $R = 8$ passed with little margin. Directly comparing the same cutoff-224 and cutoff-256 histories gave the same ordering: $R = 6$ failed, while the worst delivered-field discrepancies at $R = 8$ and $R = 10$ were respectively 7.34×10^{-4} and 1.35×10^{-4} . We therefore use the conservative value 10 rather than the merely passing value 8.

The centers are periodic and separated. Draw nonnegative simplex weights w_j with $\sum_j w_j = 1$ and set the cyclic gaps to

$$d_j = 3h + (2\pi - 3mh)w_j. \quad (10)$$

The gaps sum to 2π and are all at least $3h$. A common uniform rotation on the periodic interval then fixes the centers x_j without favoring $x = 0$.

For each requested amplitude ratio α , the code numerically solves a unit-depth permanent-form solitary wave using Tanaka's formulation [4]. Xu and Guyenne used the same solitary-wave method for their long-time finite-depth test [3]. Let Φ be the surface-potential coordinate used in the solve, X the dimensionless horizontal coordinate, $Q = e^\tau$ the fluid-speed magnitude, τ its logarithm, θ the angle of the free-surface tangent, and F_α the resulting Froude number. The code adjusts the crest value of Q by bisection until the reconstructed crest height is α , iterates Tanaka's relations, and reconstructs the dimensionless elevation $\bar{\eta}_\alpha$ from

$$\frac{dX}{d\Phi} = \frac{\cos \theta}{Q}, \quad \frac{d\bar{\eta}_\alpha}{d\Phi} = \frac{\sin \theta}{Q}.$$

It uses 257 uniformly spaced nodes on $0 \leq \gamma \leq 2.5$, reflects them about zero to obtain 513 nodes, and applies the project-fixed map $\Phi = 0.01\gamma + \gamma^5$. It performs 80 fixed-point updates. These discretization choices are ours; they are not parameters prescribed by Tanaka.

Before interpolation, the constructor verifies that both endpoint elevations and the endpoint and crest tangents differ from zero by at most 10^{-12} , then sets those quantities exactly to zero. The corrected values and tangents are joined by cubic Hermite interpolation, then extended by zero with continuous value and first derivative. Every periodic copy that intersects the domain is included. This compact extension, Hermite placement, periodization, multi-crest superposition, and spectral projection are project construction choices. In particular, a sum of profiles is an initial condition made from Tanaka solitary waves, not an exact steady multi-soliton solution. For component j ,

$$\eta_j(x) = h \sum_{r \in \mathbb{Z}} \bar{\eta}_{\alpha_j} \left(\frac{x - x_j + 2\pi r}{h} \right). \quad (11)$$

Only finitely many terms are nonzero. The interpolation is evaluated first on an eight-times finer grid and then restricted to the 1024-point grid.

Set $c_j = F_{\alpha_j} \sqrt{gh}$, with $g = 1$ in this corpus, and define

$$\rho_j(x) = (1 + \eta_{j,x}(x)^2)(c_j^2 - 2\eta_j(x)). \quad (12)$$

The following conversion of the traveling profile into the project's periodic Zakharov variables is not quoted as an equation from Tanaka. The candidate derivative of the surface potential is

$$r_j(x) = s_j [c_j - \sqrt{\rho_j(x)}].$$

A periodic antiderivative exists after subtracting the spatial mean of r_j . The program therefore defines ξ_j by

$$\partial_x \xi_j = r_j - \frac{1}{2\pi} \int_0^{2\pi} r_j(x) \, dx, \quad \int_0^{2\pi} \xi_j(x) \, dx = 0,$$

using Fourier division by ik on every nonzero mode. Finally,

$$(\eta_0, \xi_0) = P_{128} \sum_{j=1}^m (\eta_j, \xi_j).$$

This sharp projection is applied once before rollout; the result is then zero-filled into the $K = 256$ evolution band. Modes $129 \leq |k| \leq 256$ are therefore zero initially but may be generated by nonlinear evolution. A finite negative value of ρ_j is outside this construction and yields a zero-row attempt; it is never clipped to zero. A nonfinite value or an invalid wave speed signals a construction error and stops generation rather than being silently redrawn.

4.3 Benjamin–Feir carrier and sidebands

This family uses depth $h = 5$. Let ϵ_c denote the steepness of the first carrier-elevation harmonic, and let ρ denote the amplitude of each first-harmonic sideband divided by the amplitude of that carrier harmonic. Choose a carrier mode $n_c \in \{4, \dots, 20\}$ and a positive sideband separation $\Delta n < n_c$. Define the normalized sideband location

$$\beta = \frac{\Delta n / n_c}{2\sqrt{2}\epsilon_c}. \quad (13)$$

Benjamin and Feir give the historical deep-water instability condition [5]. In the present normalization, equation (4.6) of Andrade and Stuhlmeier gives $0 < \beta < 1$ [7]. Exactly 66 integer pairs $(n_c, \Delta n)$ can intersect this band for $0.05 \leq \epsilon_c \leq 0.13$; each pair is one parameter category.

Equations (5.7)–(5.11) of Andrade and Stuhlmeier give the Akhmediev-breather correspondence and the focused-envelope amplification factor $1 + 2\sqrt{1 - \beta^2}$ in this normalization. We therefore define the scalar screening proxy

$$\epsilon_f = \epsilon_c \left(1 + 2\sqrt{1 - \beta^2}\right) \quad (14)$$

and use the pre-rollout support

$$0 < \beta < 1, \quad \epsilon_f \leq F_0, \quad F_0 = \frac{1 + \sqrt{2}}{10}, \quad 0.05 \leq \rho \leq 0.10. \quad (15)$$

The value F_0 is the value of (14) at the moderate reference steepness $\epsilon_c = 0.10$ and the leading-NLS maximum-growth location $\beta = 1/\sqrt{2}$. It is our population choice, not a theorem giving a wave-breaking or regularity boundary. Yang and Liu use $\epsilon_c = 0.10$, five-percent sidebands, and the maximum-growth frequency coordinate $\Delta\omega/(\omega_c\epsilon_c) = 1$ [8, Sec. 4.3.1]. Under the leading deep-water narrowband relation $\Delta k/k_c \simeq 2\Delta\omega/\omega_c$, this corresponds to $\beta \simeq 1/\sqrt{2}$ in our symmetric-wavenumber normalization. Our sideband interval joins that five-percent value to the ten-percent perturbation of Xu and Guyenne [3].

For an elementary description of the conditional draw, set

$$\ell = \frac{\Delta n}{2\sqrt{2}n_c}, \quad \epsilon_{\min} = \max\{0.05, \ell\},$$

and

$$\epsilon_{\max} = \min\left\{0.13, \frac{-F_0 + 2\sqrt{F_0^2 + 3\ell^2}}{3}\right\}.$$

The sampler draws

$$\epsilon_c \sim \text{Unif}(\epsilon_{\min}, \epsilon_{\max}], \quad \rho \sim \text{Unif}[0.05, 0.10), \quad x_0 \sim \text{Unif}[0, 2\pi).$$

The upper endpoint is obtained by solving $\epsilon_f = F_0$ for ϵ_c at fixed $(n_c, \Delta n)$. It ranges from 0.081763 to 0.13 over the 66 categories, so no integer category is removed. The relative phase of both sidebands is fixed at $\theta = -\pi/4$. Changing the sampled support advances the Benjamin–Feir family revision from 3 to 4. The revision-4 family retains the revision-3 numerical evolution method; an old revision-3 proposal cannot be resumed into a revision-4 output root.

Because the period is 2π , set $k_c = n_c$, $k_{\pm} = n_c \pm \Delta n$, and $a = \epsilon_c/k_c$. a is the amplitude of the first carrier harmonic; ϵ_c is not computed from half the total crest-to-trough height. Let (η_c, ξ_c) be the fixed and recorded

fifth-order deep-water Stokes carrier, put $y = x - x_0$, and set the carrier phase to zero in the translated coordinate y . The initial state is

$$\begin{aligned}\eta_0(x) &= \eta_c(y) + \rho a \{ \cos(k_- y + \theta) + \cos(k_+ y + \theta) \}, \\ \xi_0(x) &= \xi_c(y) + \rho a \left\{ \sqrt{\frac{1}{k_-}} e^{k_- \eta_0(x)} \sin(k_- y + \theta) + \sqrt{\frac{1}{k_+}} e^{k_+ \eta_0(x)} \sin(k_+ y + \theta) \right\},\end{aligned}\quad (16)$$

followed by removal of the mean of ξ_0 and projection to the delivered band. The sidebands follow equation (33) of Xu and Guyenne, who in turn followed Dommermuth and Yue [3, 6]. Their single published case used

$$(n_c, \Delta n, \epsilon_c, \rho, \theta) = (9, 2, 0.13, 0.10, -\pi/4)$$

and a numerically computed steady deep-water carrier. Our code preserves their symmetric Airy-sideband formula, fixed relative phase, and use of the total η_0 inside both exponentials. Translating the complete state changes neither the relative phases nor the quartet phase $\theta + \theta = -\pi/2$. The 66 mode-pair categories, randomized parameters, finite proxy depth $h = 5$, and analytic fifth-order carrier are our generalization, so this is not an exact reproduction of their one initial state.

Let

$$T_c = \frac{2\pi}{\sqrt{k_c}}, \quad T_{\text{BF}} = 0.08 \left\lfloor \frac{100T_c}{0.08} \right\rfloor.$$

The rollout ends at the last saved-grid time not exceeding 100 linear carrier periods. Thus $100T_c$ is the intended horizon and T_{BF} is its strictly floored saved-grid endpoint. Across the 66 equally weighted mode-pair categories, T_{BF} ranges from 140.48 to 314.08 and has mean 174.50; equivalently, $T_{\text{BF}}/T_c \in [99.9541, 99.9973]$.

This support was frozen before validation stream 934 was numerically advanced. That calculation drew one case in each of the 66 mode-pair categories and evolved every case to the saved-grid endpoint immediately below its intended $100T_c$ horizon with the numerical method inherited from revision 3: the state retained Fourier modes $|k| \leq 256$, the internal Dirichlet–Neumann calculation used order four, and the time step was 0.01. All 66 attempts were accepted and none was replaced. All 13,200 retained frames were finite; the largest relative internal-Hamiltonian drift was 5.38713×10^{-5} , the largest implicit-stage residual was 9.999984×10^{-9} , and the smallest water column was 4.958158. This is evidence that the declared support is numerically usable under the tested discretization and horizon. It does not turn ϵ_f into the maximum physical slope or prove regularity of the full water-wave equations; the complete-trajectory checks in Section 5 therefore remain mandatory. Stream 934 is support-validation evidence. Because it selected cases by conditioning proposals from the preceding revision-3 law, it is not the exact-source production gate; that role belongs to revision-4 stream 935, reported in Section 9.

4.4 Relative-frequency-band JONSWAP/TMA random seas

Each random-sea parameter category fixes one depth regime, one peak-enhancement value $\gamma \in \{1, 3.3, 5\}$, and one right-moving energy fraction $r_d \in \{0, \frac{1}{2}, 1\}$. This gives $3 \cdot 3 \cdot 3 = 27$ categories.

Let $k_p > 0$ be the reference peak wavenumber and let $h > 0$ be the depth. For wavenumber $k > 0$, define

$$\omega(k, h) = \sqrt{k \tanh(kh)}, \quad \omega_p = \omega(k_p, h).$$

Revision 4 retains frequencies in the interval $\frac{1}{2}\omega_p \leq \omega \leq \frac{5}{2}\omega_p$. Phase-resolved numerical JONSWAP calculations have used exactly this interval [13], and an earlier random-wave calculation used the upper endpoint $\omega_{\text{max}} = 2.5\omega_p$ [12]. We use it as a finite component definition, not as a wave-breaking criterion. Its parameter support therefore also requires

$$\omega(128, h) \geq \frac{5}{2} \omega_p. \quad (17)$$

This ensures that the upper end of the declared relative-frequency interval fits inside the delivered band.

Let H_s denote significant wave height. In the shallow regime, choose a reference peak mode n_p uniformly from $\{16, 18, 20, 22, 24\}$. Define the relative peak depth $\chi_p = k_p h$ and the relative significant height $\delta_s = H_s/(2h)$. Propose independently

$$\chi_p \sim \text{Unif}[0.2, 1.5], \quad \delta_s \sim \text{Unif}[0.03, 0.16]$$

and repeat the pair until both $\chi_p \delta_s \leq 0.08$ and (17) hold. Equivalently, the support-admissible proposal is uniform with respect to area on the part of the rectangle satisfying both conditions; its two coordinates are not claimed to

remain independent. Set $k_p = 2\pi n_p/L$ (so $k_p = n_p$ when $L = 2\pi$), $h = \chi_p/k_p$, and $H_s = 2h\delta_s$. In the finite-depth regime, draw

$$k_p \in [2, 12], \quad h \in [0.1, 1.5], \quad H_s \in [0.005, 0.03]$$

independently and uniformly, repeating the triple until $k_p H_s/2 \leq 0.08$ and (17) both hold. The deep regime uses the same k_p and H_s ranges with $h \in [5, 25]$ and the same restriction. Thus each support-admissible proposal is uniform on the part of its original parameter box satisfying both conditions.

This bound declares the moderate input-sea regime studied here; it is not a breaking criterion. Ducroz et al. call $k_p H_s/2 = 0.08$ typical of a moderate JONSWAP sea state [15]. Their separate HOS applicability study shows why survival at a coarse spectral cutoff is not, by itself, evidence that a steep random sea is accurately resolved [14]. We therefore impose the bound before sampling phases and retain the independent whole-trajectory numerical test.

The remaining construction is deterministic once independent right- and left-moving phase arrays have been drawn uniformly on $[0, 2\pi)$. Define the peak period by $T_p = 2\pi/\omega_p$. The input k_p is a reference peak parameter; after finite-depth modification, cell integration, and the relative-frequency restriction, the largest discrete cell need not be centered exactly at k_p . Define $\sigma = 0.07$ for $\omega \leq \omega_p$ and $\sigma = 0.09$ for $\omega > \omega_p$. The unnormalized JONSWAP frequency density is [10]

$$\tilde{S}_J(\omega) = g^2 \omega^{-5} \exp\left[-\frac{5}{4} \left(\frac{\omega_p}{\omega}\right)^4\right] \gamma^{\exp[-(\omega-\omega_p)^2/(2\sigma^2\omega_p^2)]}. \quad (18)$$

The factor $g^2 = 1$ could be omitted numerically because the density is normalized below, but it is retained here to display the published spectral shape. The TMA finite-depth correction of Bouws et al. [11] is

$$\Phi(kh) = \frac{\tanh^2(kh)}{1 + 2kh/\sinh(2kh)}. \quad (19)$$

Thus the frequency density used is $\tilde{S}_J(\omega)\Phi(kh)$. This reduces low-frequency energy in shallow water and tends to the ordinary JONSWAP form in deep water.

Let $\Delta k = 2\pi/L = 1$, $K = 128$, and $k_j = j\Delta k$. For each positive mode $j \leq 128$, define the clipped Fourier-cell endpoints

$$a_j = \max\{0, k_j - \Delta k/2\}, \quad b_j = \min\{K, k_j + \Delta k/2\}.$$

Define the group velocity $c_g(k, h) = \partial\omega(k, h)/\partial k$. The program uses fixed 16-point Gauss–Legendre quadrature and the sharp relative-frequency indicator

$$I_p(k) = \begin{cases} 1, & \frac{1}{2} \leq \frac{\omega(k, h)}{\omega_p} \leq \frac{5}{2}, \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

to compute the nonnegative cell energy

$$E_j = \int_{a_j}^{b_j} \tilde{S}_J(\omega(k, h)) \Phi(kh) c_g(k, h) I_p(k) dk.$$

This is the frequency density converted to a wavenumber density and integrated over one Fourier cell. The indicator is evaluated at the quadrature nodes, so a boundary cell can be only partly retained. Normalize the cell energies by $w_j = E_j / \sum_{\ell=1}^{128} E_\ell$. The prescribed surface-elevation variance is $m_0 = (H_s/4)^2$. Set

$$A_{j,+} = \sqrt{2r_d m_0 w_j}, \quad A_{j,-} = \sqrt{2(1-r_d) m_0 w_j}.$$

For the stored phases $\phi_{j,+}$ and $\phi_{j,-}$, the initial state is

$$\begin{aligned} \eta_0(x) &= \sum_{j=1}^{128} \{A_{j,+} \cos(jx + \phi_{j,+}) + A_{j,-} \cos(jx + \phi_{j,-})\}, \\ \xi_0(x) &= \sum_{j=1}^{128} \frac{\omega(j, h)}{j \tanh(jh)} \{A_{j,+} \sin(jx + \phi_{j,+}) - A_{j,-} \sin(jx + \phi_{j,-})\}. \end{aligned} \quad (21)$$

The constructed state is projected once to its declared spectral band. A realization enters the rollout only if every value is finite and

$$\min_{\ell} \{h + \eta_0(x_\ell)\} > 0.$$

This is the elementary graph-domain condition: the free surface must lie strictly above the flat bottom. If one realization in a batch fails it, only that case is recorded as a zero-row attempt and replaced in its own parameter category; its valid batch companions are retained.

For JONSWAP/TMA, the family event in the common accepted-case law below specifically includes the phase-dependent initial construction and graph condition, followed by the adjustment and autonomous trajectory decisions in Section 5. This is not a separate phase-valued support gate: it is the common complete-case conditioning specialized to this family. The JONSWAP/TMA completion audit reports its effect separately; the common-law statement does not claim measured bias in another family. No retrospective parameter cutoff or bulk rejection-rate threshold is introduced.

For each valid realization the record contains the discrete peak wavenumber, the number of spectral cells at least half the peak cell energy, the root mean squares of η_0 and ξ_0 , the realized-to-requested significant height ratio, the relative error in the linear Hamiltonian implied by the sampled cell energies, and $\min_{\ell}\{h + \eta_0(x_{\ell})\}$. These values diagnose the discrete construction; none is an additional acceptance threshold. No realization-dependent rescaling or visual roughness test is applied. Hasselmann et al. supply the continuous JONSWAP form [10], and Bouws et al. supply the TMA factor [11]. Draycott et al. use the complete interval $0.5\omega_p \leq \omega \leq 2.5\omega_p$ [13], while Grice et al. provide separate precedent for its upper endpoint $2.5\omega_p$ [12]. The parameter strata, $\gamma \in \{1, 3.3, 5\}$, direction split, independent phases, and Fourier-cell quadrature remain project choices rather than claims from those sources.

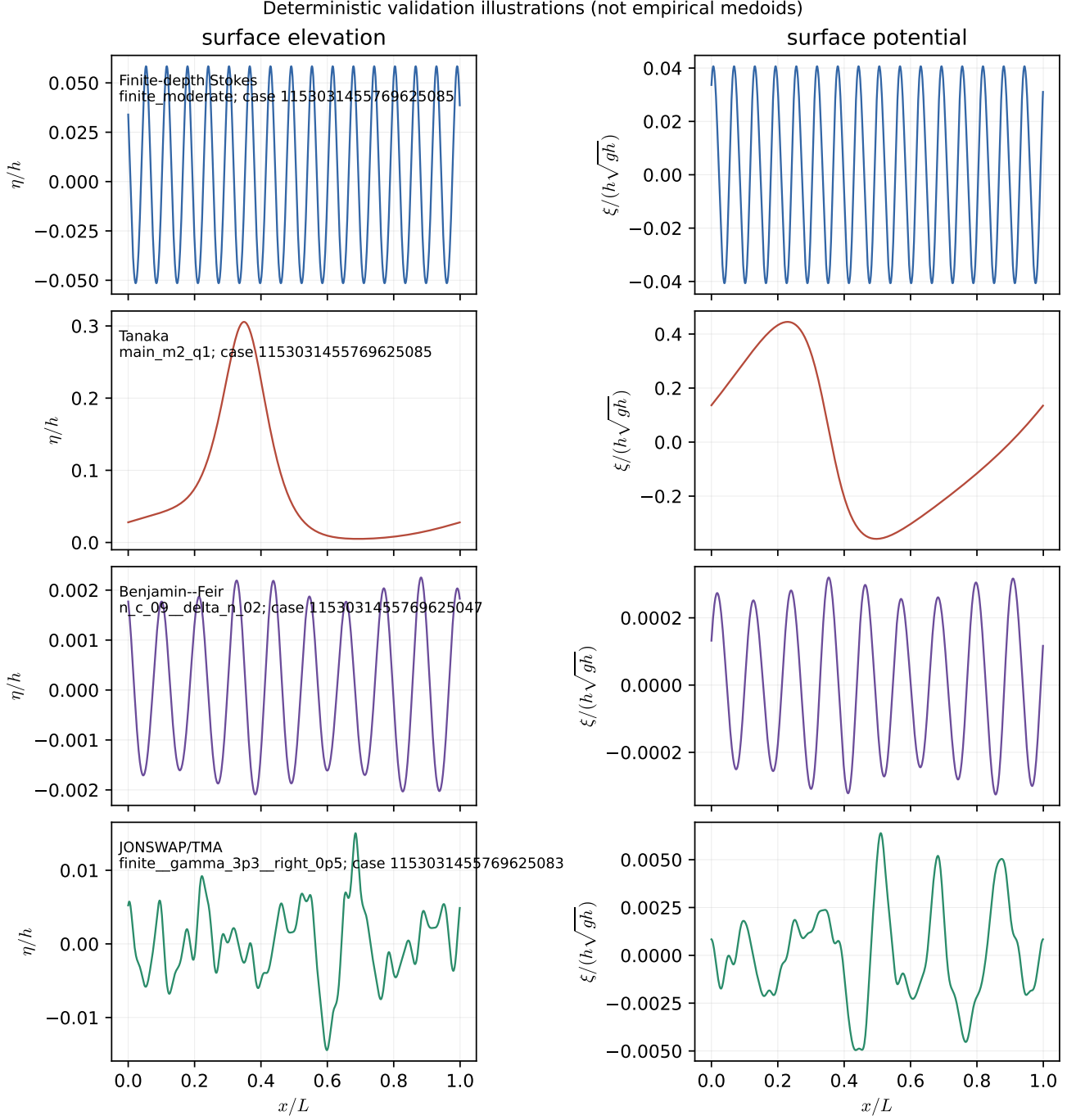


Figure 1: Deterministic illustrations from accepted validation cases in one declared central category per family, selected by the lower-median case ID; they are not empirical medoids. The rows are Stokes revision 2 case 1153031455769625085 (`finite_moderate`), Tanaka revision 3 case 1153031455769625085 (`main_m2_q1`), Benjamin–Feir revision 4 case 1153031455769625047 (`n_c_09_delta_n_02`), and JONSWAP/TMA revision 4 case 1153031455769625083 (`finite_gamma_3p3_right_0p5`). Each panel shows the first retained row; the left column is η/h and the right column is $\xi/(h\sqrt{gh})$, with $g = 1$.

5 One numerical rollout and one whole-trajectory decision

Stokes cases are static. Let F denote one of the three trajectory families. The current internal grid size, cutoff, and DNO-series order are

$$(N_F, K_F) = \begin{cases} (1024, 256), & F = \text{Tanaka or Benjamin-Feir}, \\ (2048, 704), & F = \text{JONSWAP/TMA}, \end{cases}$$

and

$$M_F = \begin{cases} 6, & F = \text{Tanaka}, \\ 4, & F = \text{Benjamin-Feir or JONSWAP/TMA}. \end{cases}$$

This definition applies only to the three rollout families. Static Stokes has no evolution cutoff because it is not time-integrated; its single stored label nevertheless uses the common delivered P_{128} target in (3). The unprojected internal DNO output is

$$\tilde{G}_F(\eta; h)\xi = \mathcal{G}_{M_F}^{N_F, 8}(P_{K_F}\eta; h)(P_{K_F}\xi), \quad G_{\text{evol}, F}(\eta; h)\xi = P_{K_F}^\circ \left[\tilde{G}_F(\eta; h)\xi \right].$$

The discrete Zakharov equations used for the rollout are

$$\begin{aligned} \eta_t &= G_{\text{evol}, F}(\eta; h)\xi, \\ \xi_t &= P_{K_F}^\circ \left[-\eta - \frac{1}{2}\xi_x^2 + \frac{(\tilde{G}_F(\eta; h)\xi + \eta_x \xi_x)^2}{2(1 + \eta_x^2)} \right]. \end{aligned} \quad (22)$$

The Zakharov surface formulation and the Craig–Sulem expansion follow the formulation used by Xu and Guyenne [3]. The truncation order, spatial grid, padding factor, and delivered Fourier band in this note are fixed project choices. The Craig–Sulem calculation inside $\tilde{G}_F$ uses padding factor eight. Its unprojected value enters the quadratic term before the complete ξ_t right-hand side is projected. The products written explicitly in the second line of (22) are formed on a separate two-times Fourier grid before projection.

The linear gravity-wave part of (22) is advanced exactly in Fourier space. More precisely, write the right-hand side as $L_h z + R_h(z)$, where $z = (\eta, \xi)$, $L_h z$ is the linear part, and $R_h(z)$ is the nonlinear remainder. A Benjamin–Feir case starts this autonomous equation immediately. A JONSWAP/TMA case first solves the Dommermuth-adjusted equation [9]

$$z_t = L_h z + A(t)R_h(z), \quad A(t) = 1 - \exp \left[- \left(\frac{t}{10T_p} \right)^4 \right]. \quad (23)$$

Dommermuth supplies the ramped-nonlinearity construction; the exponent four, the scale $10T_p$, and the $20T_p$ handoff horizon are fixed project settings validated below rather than unique prescriptions of that method. until

$$T_{\text{adj}} = 0.08 \left\lfloor \frac{20T_p}{0.08} \right\rfloor. \quad (24)$$

This is the last saved-grid time not exceeding $20T_p$. The complete 2048-point, $K_F = 704$ endpoint is handed to the autonomous equation without projection to P_{128} and without rescaling either component. At that handoff the production clock is restarted at zero and A is fixed to one.

For either equation, the change of variable $v(t) = e^{-tL_h} z(t)$ gives

$$v_t = e^{-tL_h} R_h(e^{tL_h} v).$$

During adjustment, the right-hand side is additionally multiplied by $A(t)$. The program applies the two-stage, fourth-order Gauss–Legendre method (GL2) to this transformed equation. This is the integrating-factor GL2 method described by Xu and Guyenne [3]. The step, iteration cap, and residual test below are our fixed implementation settings.

The GL2 step is 0.01. At each step it solves two coupled internal stage equations. JONSWAP/TMA permits at most five simultaneous fixed-point updates; Tanaka and Benjamin–Feir permit at most four. For each stage and each field separately, the remaining stage-equation mismatch is divided by the larger of the stage norm, the fixed-point-map norm, and the smallest normal floating-point number. The largest of these relative mismatches must be at most 10^{-8} . After each Tanaka step, the program multiplies every retained Fourier coefficient of both state fields by

$$\sigma_k = \exp \left[-36 \left(\frac{|k|}{256} \right)^{36} \right]. \quad (25)$$

Benjamin–Feir uses the sharp cutoff $|k| \leq 256$, while JONSWAP/TMA uses the sharp cutoff $|k| \leq 704$ on 2048 points. After each completed GL2 step, both state fields are sharply projected to P_{256} or P_{704} , respectively. These repeated projections enforce the declared Galerkin space, but they are not Hou–Li filters and do not smoothly damp the retained modes. A candidate state is stored every 0.08, exactly every eight steps; there is no time interpolation. Every delivered state is projected to P_{128} , and its label is then evaluated by the common order-six formula (3). Thus modes above 128 and, for Benjamin–Feir and JONSWAP/TMA, the order-four DNO are numerical workspace for evolution, not additional model inputs or outputs. For Tanaka, the exact revision-3 chain is: construct and sharply project the initial state to P_{128} , zero-fill it into $K_F = 256$, apply (25) to both state fields after every GL2 step, and expose only the P_{128} state and common P_{128} target. Benjamin–Feir and JONSWAP/TMA use the same delivered band but their distinct sharply projected, order-four internal grids and cutoffs stated above.

The step 0.01 was selected before production on a fixed historical revision-2 12-case panel. All ten cases that completed at both steps 0.01 and 0.005 agreed to at most 1.126337×10^{-5} ; the other two became incomplete under both steps. The 0.005 calculation is retained as method-level time-step evidence, not as a second rollout applied to each production case or as current-revision population evidence for any family.

Tanaka requests $0 \leq t \leq 200$. Benjamin–Feir requests the case-dependent time T_{BF} defined above. After adjustment, a random sea requests the last 0.08-grid time no later than $16T_p$. Its autonomous time coordinate starts from zero at the full-band handoff. Cases with different random-sea horizons may share a numerical batch, but each one is judged only on its own requested prefix.

Suppose the initial condition already obeys its family’s parameter rules. There is then one trajectory decision:

$$\boxed{\text{accept} \iff \text{the fixed method returns one complete admissible trajectory.}} \quad (26)$$

Here *complete admissible trajectory* means that the calculation reaches its requested final time, every GL2 stage has finite relative residual at most 10^{-8} , every stage and updated state is finite, all saved η , ξ , and $G_{\text{ref}}(\eta; h)\xi$ values are finite, and

$$h + \eta(x_\ell, t_j) > 0$$

at every saved time t_j and grid point x_ℓ . The final inequality is the elementary geometric requirement that the graph of the free surface remain above the flat bottom $y = -h$. It has already been checked at $t = 0$ while assembling the initial batch and is checked again on every saved state. The recorded diagnostics state which clause failed; they are not additional fitted filters. A failed trajectory contributes zero rows. Keeping its finite prefix would systematically give difficult cases only their earlier, easier states.

Benjamin–Feir revision 4 and JONSWAP/TMA revision 4 cases have one additional numerical condition. On every saved frame of their autonomous trajectory, before projection to P_{128} , use $K_F = 256$ for Benjamin–Feir and $K_F = 704$ for JONSWAP/TMA and define

$$H_F(t_j) = \frac{1}{2} \int_0^{2\pi} \left\{ \xi G_4^{K_F}(\eta; h)\xi + g\eta^2 \right\} dx, \quad d_{H,F} = \max_j \frac{|H_F(t_j) - H_F(0)|}{\max\{|H_F(0)|, \varepsilon_{64}\}}, \quad (27)$$

where $G_4^{K_F}$ is the order-four DNO evaluated from the full sharp internal state and ε_{64} is the smallest positive normal float64 number. These cases require

$$d_{H,F} \leq 10^{-3}. \quad (28)$$

At the same saved frames the executor records whether the internal state and $G_4^{K_F}(\eta; h)\xi$ are finite and records the minimum value of $h + \eta$. It persists these indicators, the initial Hamiltonian, the maximum drift, and the minimum water column with the case. For JONSWAP/TMA, $H_F(0)$ is computed after the full-band handoff. Equation (28) is not imposed during (23), because the ramp makes that equation explicitly time dependent; finiteness, positive water depth, and solved implicit stages are still required during adjustment.

Only the declared $\Delta t = 0.01$ path is run for each production case. A failure owns zero rows and is replaced from the same parameter category. Sign changes, slope, extrema, spectrum, absolute target size, visual appearance, and model error do not decide acceptance. Hamiltonian drift decides acceptance only through (28) for the two stated families.

A static Stokes state has no trajectory. It is kept only when its sampled parameters obey the Stokes rules, its state and common target are finite, and $h + \eta_0 > 0$ at every grid point.

5.1 Why the JONSWAP/TMA cutoff is 704

The predeclared shallow sentinel used the earlier revision-3 spectral constructor and was adjusted and evolved independently at two sharp cutoffs on the 2048-point grid. The $K = 704$ and $K = 768$ arms used the numerical

method inherited by revision 4. Over the 16-peak-period autonomous history, their largest relative whole-field differences were

$$(e_\eta, e_\xi, e_{G\xi}) = (4.64267, 1.93035, 7.91931) \times 10^{-4}. \quad (29)$$

The target in this comparison was recomputed after the normalized transfer in (5); the internal order-four DNO output was not resampled. All three values in (29) are below 10^{-3} . The corresponding $K = 640$ versus $K = 768$ comparison exceeded 10^{-3} , so 640 is not the production cutoff. Diagnostics formed only from modes $96 \leq |k| \leq 128$ were about 3×10^{-3} . The conclusion is therefore only a global whole-field statement, not a bandwise convergence statement. Some stage residuals are close to the required 10^{-8} , and this is one worst-case sentinel. It therefore did not replace a gate using the current constructor; the revision-4 gate has since passed as reported in Section 9.

The then-current revision-3 source was replayed from the archived full internal $K = 704$ endpoint. It reproduced all 289 autonomous saved frames with maximum relative L^2 errors ($3.99 \times 10^{-16}, 4.66 \times 10^{-16}, 1.17 \times 10^{-15}$) in $(\eta, \xi, G(\eta)\xi)$. The replay was accepted, solved every stage, had maximum residual 9.994198×10^{-9} , Hamiltonian drift 1.831187×10^{-5} , minimum water column 0.0387699, and CPU wall time 197.355 seconds. This verifies that historical source and the inherited evolution method against the archived sentinel; it does not validate the revision-4 relative-frequency population.

5.2 Endpoint diagnostics that do not reject cases

Spatial roughness and sign changes were used historically to find cases that merited inspection; neither quantity decides acceptance. We applied two such rankings to the revision-2 method-level rollout panel. We first removed the two trajectories that did not reach their requested terminal time, exactly as the corpus rule requires. For the figures below, we then omit the three JONSWAP/TMA cases because their intentionally broadband spectra otherwise dominate both rankings and obscure comparison among the coherent-wave families. This second omission is only a presentation choice; it does not remove any case from the corpus. The displayed rankings therefore use five completed Tanaka and Benjamin–Feir endpoints. No last-finite frame from an incomplete trajectory participates.

Let $\hat{\eta}_k$ be the Fourier coefficients of an endpoint elevation after removal of its mean. Define its root-mean-square wavenumber by

$$k_{\text{eff}}(\eta) = \left(\frac{\sum_{k \neq 0} k^2 |\hat{\eta}_k|^2}{\sum_{k \neq 0} |\hat{\eta}_k|^2} \right)^{1/2}. \quad (30)$$

This is large when a substantial fraction of the elevation energy lies at short spatial scales; by convention it is zero when the denominator vanishes. Figure 2 shows all five completed non-JONSWAP/TMA cases in decreasing order.

For the second ranking, let $f_\ell = [G_{\text{ref}}(\eta; h)\xi](x_\ell)$ at the endpoint and form the cyclic forward differences

$$d_\ell = f_{\ell+1} - f_\ell, \quad \ell = 0, \dots, 1023, \quad f_{1024} = f_0.$$

Set $\tau = 0.03 \max_\ell |d_\ell|$, discard differences with $|d_\ell| \leq \tau$, and count cyclic transitions between the signs of the remaining differences. Denote this count by $C_{0.03}(G_{\text{ref}}(\eta; h)\xi)$. The dead zone prevents roundoff-sized differences from creating sign changes. Figure 3 shows all five completed non-JONSWAP/TMA cases.

Completed non-JONSWAP endpoints: diagnostic rankings, not rejection rules

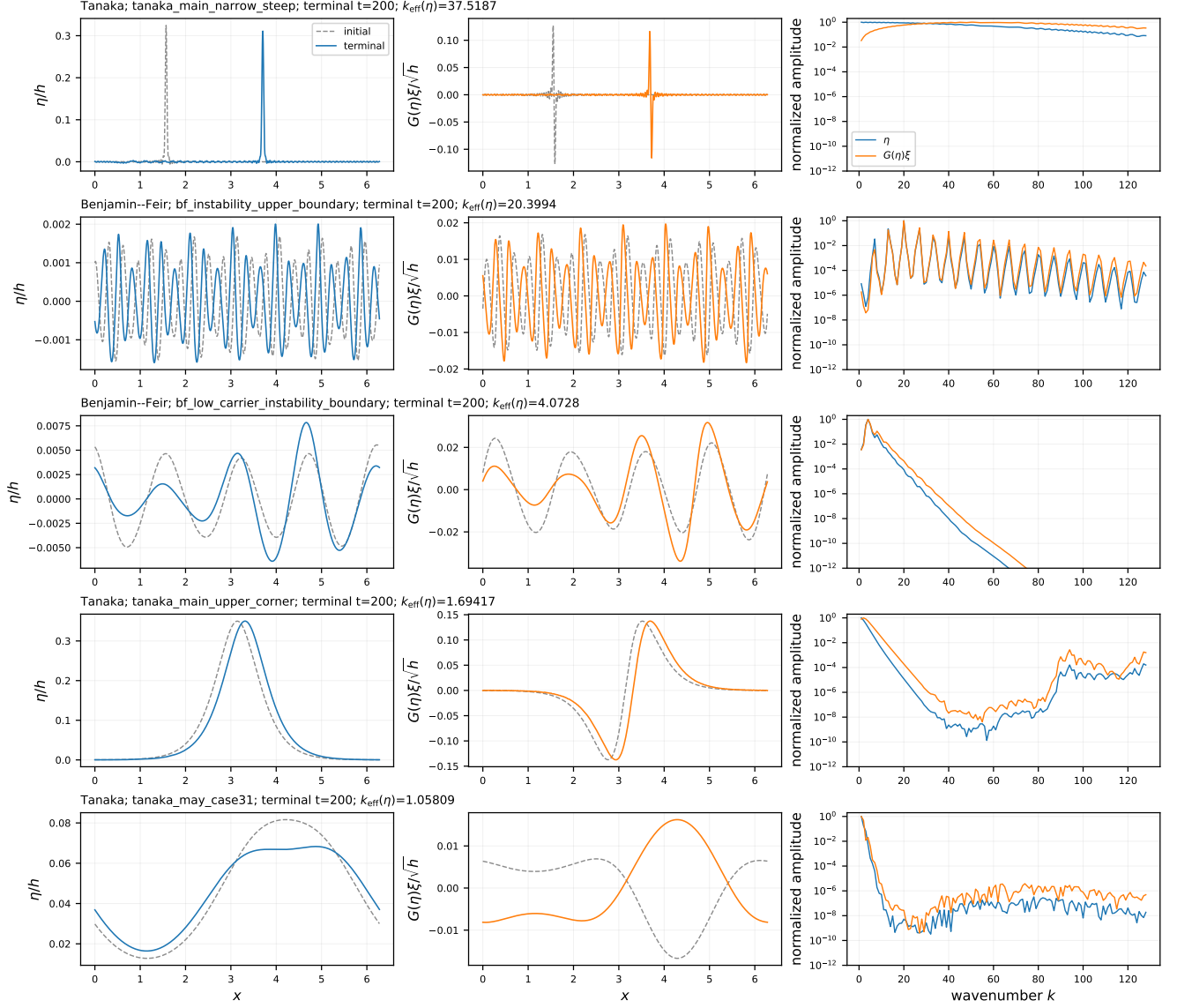


Figure 2: Historical revision-2 method-panel cases with the largest endpoint value of $k_{\text{eff}}(\eta)$. In each row, the dashed gray curve is the initial field, the colored curve is the terminal field, and the right panel shows the corresponding normalized Fourier amplitudes. Every displayed trajectory completed its requested interval, and JONSWAP/TMA cases are omitted from this display. The narrow Tanaka case has the largest value, $k_{\text{eff}} = 37.52$; spatial roughness alone is therefore not evidence of numerical failure.

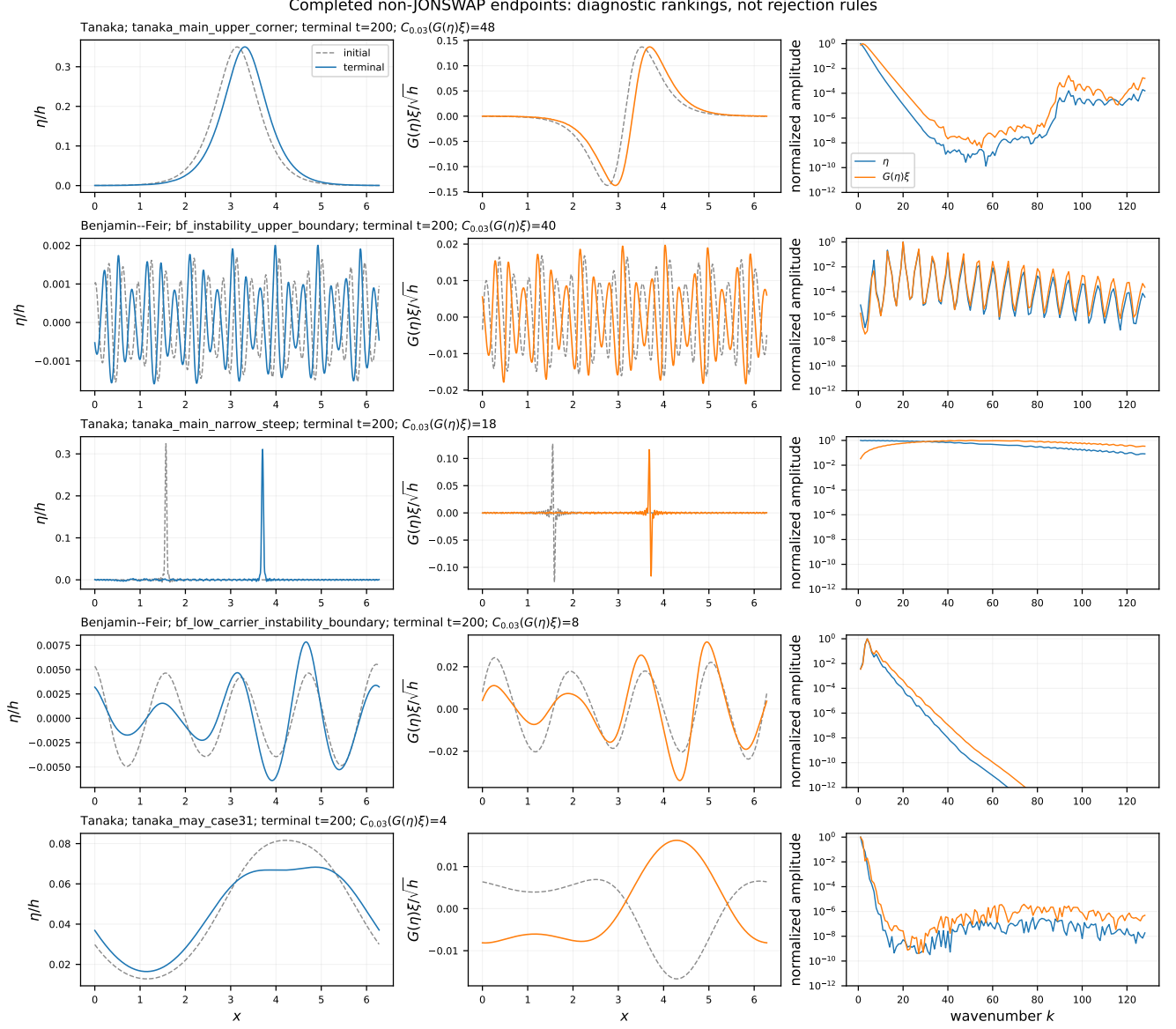


Figure 3: Historical revision-2 method-panel cases with the largest endpoint sign-change count in the cyclic forward difference of $G_{\text{ref}}(\eta; h)\xi$, after incomplete trajectories and the broadband JONSWAP/TMA display cases have been removed. The completed Tanaka upper-corner and Benjamin-Feir upper-boundary cases lead with 48 and 40 changes, followed by the narrow Tanaka case with 18. The count detects small oscillatory contributions but does not by itself show that a completed trajectory is numerically unstable, so it is retained only as a diagnostic.

6 Selecting times after acceptance

Time selection reduces storage; it never decides whether a case is accepted and it never uses model error. Stokes retains only $t = 0$. Tanaka has 2501 candidate times

$$t_j = 0.08j, \quad j = 0, \dots, 2500,$$

and retains exactly 200 distinct times, including both endpoints.

For Tanaka, define the total surface-slope energy

$$S_{T,j} = \sum_{\ell=0}^{1023} |\eta_x(x_\ell, t_j)|^2$$

and its discrete time difference

$$DS_{T,j} = \begin{cases} S_{T,1} - S_{T,0}, & j = 0, \\ (S_{T,j+1} - S_{T,j-1})/2, & 1 \leq j \leq 2499, \\ S_{T,2500} - S_{T,2499}, & j = 2500. \end{cases}$$

The Tanaka activity is

$$A_{T,j} = \frac{|DS_{T,j}|}{S_{T,j} + 10^{-12}}.$$

Thus extra weight is given to times when the surface shape changes rapidly.

The Tanaka activity is smoothed by a fixed Gaussian in the time index. Its standard deviation is 50 candidate times. The kernel stops at three standard deviations, and values beyond either endpoint are replaced by that endpoint value. If $\tilde{A}_{T,j}$ is the smoothed activity, the selection weights are

$$p_{T,j} = \frac{1 - \alpha_T}{2501} + \alpha_T \frac{\tilde{A}_{T,j}}{\sum_{r=0}^{2500} \tilde{A}_{T,r}}, \quad \alpha_T = 0.50.$$

If all activities vanish, the second term is replaced by a uniform weight. The program takes fixed midpoint quantiles of the cumulative weights, pins the first and last times, and clips each quantile index so that enough unused indices remain for all later quantiles. A forward pass then advances any repeated index to the next unused one. The result is deterministic.

For Benjamin–Feir, let $J_{BF} = T_{BF}/0.08$. Store the 200 indices

$$j_\ell = \left\lfloor \frac{\ell J_{BF}}{199} + \frac{1}{2} \right\rfloor, \quad \ell = 0, \dots, 199.$$

These indices are approximately uniform in time and include both endpoints. Thus the storage rule introduces no additional event score beyond the declared 100-carrier-period horizon.

For a random sea, let

$$T_s = 0.08 \left\lfloor \frac{16T_p}{0.08} \right\rfloor, \quad J_s = T_s/0.08.$$

It retains the 16 indices

$$j_\ell = \left\lfloor \frac{\ell J_s}{15} + \frac{1}{2} \right\rfloor, \quad \ell = 0, \dots, 15.$$

These are approximately uniform and include 0 and T_s . Here $t = 0$ is the adjusted full-band endpoint and the start of autonomous production, not the initial linear random sea.

7 Identity, splitting, quotas, and storage

7.1 Case identity and data splits

The full case identity consists of

(family, revision, split, stream, attempt).

Its deterministic random stream instead uses these five integer seed words:

(split root, family, revision, stream, attempt).

These words seed NumPy’s PCG64 generator. The split roots are

split	training	validation	test
root	2026072210	2026072204	2026072205.

The split, parameter category, and entire sampled specification are written before construction or integration. A rejection remains attached to that split. All stored times from one trajectory therefore stay together, so nearby times from the same numerical solution cannot cross into validation or test.

7.2 Accepted-case quotas

A quota is a required number of accepted complete cases, not a number of attempts or stored rows. Suppose a family has m ordered parameter categories and requires C accepted cases. Write

$$C = bm + r, \quad 0 \leq r < m.$$

The first r categories receive $b + 1$ cases and the other categories receive b . This is the most even integer division possible.

A failed attempt is replaced only in its original category. A category assigned Q accepted cases may use at most $4Q$ recorded attempts. If the resolved attempts reach this bound without filling Q , the family run stops as incomplete. Another category cannot fill the deficit. The factor four only bounds computation; it does not change any case-level decision. The Stokes amplitude redraws described above are internal to one attempt.

The family sampling formulas govern attempted specifications, not the unconditional law of loader-visible cases. If X is a complete attempted specification in preassigned category i , A_f is the declared complete case-acceptance event for family f , and w_i is the accepted-quota weight, then

$$p_{\text{release}}(i, x) = w_i p_{\text{proposal}}(x \mid i, A_f = 1).$$

For Stokes, A_f comprises declared static support, a finite state, positive water column, and a finite target. For Tanaka, it comprises declared construction support followed by complete autonomous-trajectory acceptance; for Benjamin–Feir, initial-state validity followed by complete autonomous- trajectory acceptance. For JONSWAP/TMA, it comprises initial construction and graph validity followed by nonlinear adjustment and complete autonomous-trajectory acceptance. Consequently the quota preserves the declared category marginals, not the unconditioned proposal law within a category. This identity describes release construction; it does not by itself claim a measured within-category shift for every family. Every rejected proposal remains recorded, and no retrospective parameter cutoff is applied.

7.3 Files and combined view

An accepted row stores three length-1024 spatial arrays η , ξ , and $G_{\text{ref}}(\eta; h)\xi$ in float32, together with float64 depth and time, its selected candidate-time index, and its case-local index in that array file. The case record preserves the sampled parameters and numerical decision for both accepted and rejected attempts. It also binds the family, split, random stream, numerical settings, software versions, source hashes, and array checksum.

After the four families have been generated for all three splits, the combined builder checks the records, shapes, checksums, and required counts. It allows exactly one revision for each family. Source and numerical settings must agree among all chunks and splits with the same (family, revision) pair; settings from two different families are not required to be identical. This is necessary because Tanaka revision 3 uses 1024 points and internal cutoff 256, Benjamin–Feir revision 4 uses the same grid and cutoff with a different DNO order, JONSWAP/TMA revision 4 uses 2048 points and cutoff 704, and revision-2 Stokes is static. The builder then writes a compact row-to-case index pointing to the existing arrays rather than copying them. This index is the dataset view read by the trainer.

8 Production size and trainer handoff

The first checkpoint requests, from *each* family,

2048 training, 1024 validation, 1024 test cases.

The four families therefore contain 16,384 accepted cases in total. For one accepted case from each family, the stored-row count is

$$1 + 200 + 200 + 16 = 417.$$

The exact split counts are The authenticated final view references 1315 physical field shards containing 7686144

Split	Accepted cases	Stored rows
Training	8,192	854,016
Validation	4,096	427,008
Test	4,096	427,008
Total	16,384	1,708,032

rows and 94,666,557,624 measured bytes (88.165102 GiB). Cumulative views reference immutable source prefixes and do not copy those fields. Training then grows by additive, disjoint chunks to

$$C_{\text{tr}} \in \{2048, 4096, 8192, 16384\}$$

accepted cases per family; validation and test remain fixed at 1024 cases per family. The resulting cumulative sizes are

Training cases per family	Total stored rows
2,048	1,708,032
4,096	2,562,048
8,192	4,270,080
16,384	7,686,144

Thus 2048 is a checkpoint, not the completed corpus. The target at 16,384 contains 73,728 accepted cases across the four families and three splits. Its 7,686,144 rows are close to the 7,633,989 rows in historical v9: 52,155 more rows, or 0.68%.

Normalization is computed from all stored *training* rows and from no validation, test, or diagnostic rows. Let η_i , ξ_i , and h_i denote the elevation, potential, and depth in training row i , and let $G_i^* = G_{\text{ref}}(\eta_i; h_i)\xi_i$ denote its reference label. Define

$$s_\eta = \max_{i,x} |\eta_i(x)|, \quad s_\xi = \max_{i,x} |\xi_i(x)|, \quad s_G = \max_{i,x} |G_i^*(x)|.$$

The authenticated training-only values are $s_\eta = 0.15583430230617523$, $s_\xi = 0.09394174814224243$, and $s_G = 0.12198150902986526$, computed from 6832128 rows. The release audit verifies all three before training. The current loader uses one as a defensive fallback if a nonpositive scale reaches it; that fallback is not treated as validation of a malformed corpus. The cached values are tied to the checked dataset identity; their statistics fingerprint is 4e8ec96947bc05cee3b375467499b0ae00a7814f9982614084efd859efe7c3b8. The trainer receives

$$\eta/s_\eta, \quad \xi/s_\xi, \quad \log(h/h_\star), \quad G_{\text{ref}}(\eta; h)\xi/s_G,$$

where the reference length is $h_\star = L/(2\pi) = 1$ on the production domain. The paper-corpus training command must explicitly select this scale normalization with **-norm scale**; the trainer’s generic min–max default is not the rule stated here.

At every training epoch, the loader chooses one stored time uniformly from every accepted training case. Thus each family contributes C_{tr} samples per epoch even though the archive stores different numbers of times per case. Validation uses one fixed deterministic time per case. Airy, manufactured-DNO, and sparse-Fourier states remain diagnostic tests and do not enter training or its normalization.

9 Checks before bulk generation

Grid, truncation, padding, spectral support, and time step are chosen on fixed method-level panels before bulk generation. They are not re-estimated for individual production cases. The current family-specific checks are not one mixed numerical batch. The Tanaka revision-3 pilot used complementary six- and five-case shards and passed: all 11 attempted cases were accepted, none was rejected, and the accepted trajectories produced 2200 rows. The artifacts carry validation split tags so that the transaction path is tested, but their dedicated pilot roots are

diagnostic and are excluded from final validation data, training, and normalization. This establishes the eleven Tanaka categories only; it is not part of corrected bulk production.

The revision-4 Benjamin–Feir production-law gate used stream 935. It accepted all 66 first proposals, rejected none, and retained 13,200 rows. Over its accepted cases, the largest implicit-stage residual was 9.999967×10^{-9} , the largest relative internal-Hamiltonian drift was 2.952009×10^{-5} , and the smallest water column was 4.960598. The earlier stream-934 result remains independent support-validation evidence; stream 935 is the direct revision-4 production gate.

A preceding revision-3 JONSWAP/TMA source gate used stream 933. It accepted one case in every category after 29 attempts, retained 432 rows, and rejected two attempts with zero rows. The first proposals in `finite__gamma_3p3__right_1` and `shallow__gamma_1__right_0p5` stopped during nonlinear adjustment; the canonical second proposal in each category passed. Among the 27 accepted cases, the corresponding three extrema were 9.999783×10^{-9} , 5.908800×10^{-5} , and 0.0127322. This gate remains historical evidence for the inherited adjustment and evolution method; it does not test the revision-4 relative-frequency constructor.

The independent audit rederived every proposal and run specification and passed. Its artifact is `outputs/noncorpus_bf_revision4_jonswap_revision3_category_smoke_20260802/category_smoke_release_audit_v1.json`, with SHA-256

62fb5fe3b6be7df0d21784b1ce590c44cc484bf5fdf84ad6097fbed6f655f684. This preserves the independent audit of the revision-4 Benjamin–Feir gate and the historical revision-3 JONSWAP/TMA gate; one case per category does not estimate a population rejection rate. The JONSWAP/TMA $K = 704$ versus $K = 768$ sentinel remains the separate whole-field cutoff check in (29). Likewise, the four static Stokes revision-2 categories passed their own exact calculation, with all four cases accepted and all failed-quality masks zero. One case per category tests construction and evolution.

The direct revision-4 JONSWAP/TMA gate fixed 20 accepted cases in each of the 27 categories before execution. It retained 540 cases from 548 attempts; eight incomplete numerical attempts retained zero rows. The observed rejection rate was 1.46%, below the predeclared two-percent bound. This is the current-constructor gate used before revision-4 bulk generation, not a bulk release threshold. Its audited summary is `outputs/jonswap_relative_band_fresh_gate_20260804/jonswap_relative_band_fresh_gate.summary.json`.

A family’s bulk generation began only after its own current-source check passed. The final release audit then checked exact case and row counts, disjoint splits, finite stored arrays, positive depths, zero-row failed cases, checksums, and training-only normalization.

Final authenticated status. All four family revisions are complete at 16384 training, 1024 validation, and 1024 test cases each. Stokes retained 18432 cases from 18432 attempts and stored 18432 rows; corrected Tanaka retained 18432/18432 and stored 3686400 rows; Benjamin–Feir retained 18432/18455 and stored 3686400 rows; and JONSWAP/TMA retained 18432/18726 and stored 294912 rows. Thus the release has 73728 accepted cases from 74045 attempts, 317 zero-row rejections, and 7686144 rows. The combined rejection rate is 0.428118%. The historical 4096-case Tanaka population and all diagnostic gates remain outside the release.

The cumulative combined-summary SHA-256 identities for 2048, 4096, 8192, and 16384 training cases per family are, respectively:

```
2265cb71270accb3f6d9103e90433b664ca813989dfab63403d8612af22d28c3
7feb018ee4d94eab10278917c145262026ccab6169eaaa2d8877f7c7b28c9da
43b1eb91198e724c6f32b871c2d0efd0d8355533789919df5897ad588e7e85f6
32f764cf865c90892ee0329e48fe992cbc24df2707e05fd1af2ee9cd2b84307c.
```

The final 26-source manifest and trajectory-map SHA-256 identities are

```
dbb76ed1cba4d0146f3c9643a73d168d3a9fa97bc27c37fe0fd7ad08554bd734
f26888ad0a2c738405fb1662b9410337e1babfe48f7913805b5243ae0f279cc8.
```

The immutable document handoff has SHA-256

```
b21830cf56f9c17a1ee165624c71fdc0e286727747d364330fbace19e5ef6053.
```

No reported model has been trained on this replacement dataset.

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